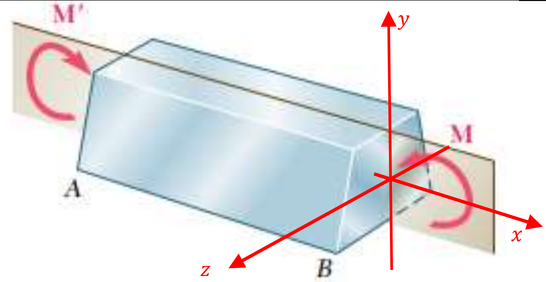
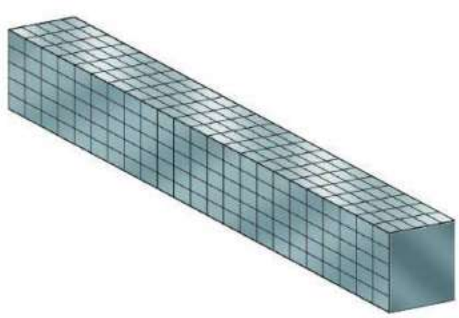
Symmetric bending

Normal stresses in bending

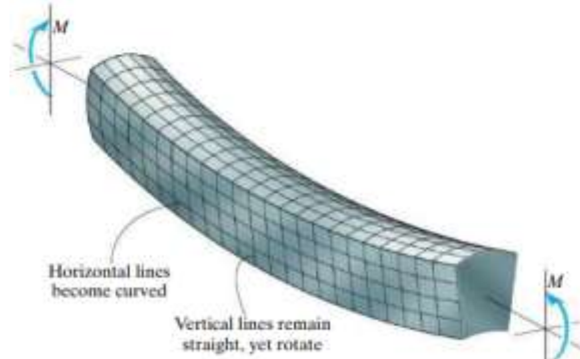
Consider a straight prismatic beam made of homogeneous material, having a cross-sectional area that is symmetrical with respect to an axis, and the bending moment is applied about an axis perpendicular to this axis of symmetry (principal axes) as shown. If the bending moment is constant, it is called pure bending (simple or symmetric bending)



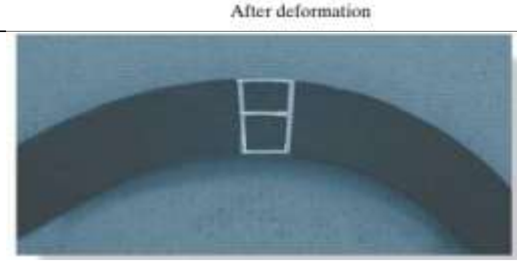
By using a highly deformable material such as rubber, we can illustrate what happens when a straight prismatic member is subjected to a bending moment. Consider the undeformed bar which has a square cross section and is marked with longitudinal and transverse grid lines. When a bending moment is applied, it tends to distort these lines into the pattern shown. Notice that the longitudinal lines become curved and the vertical transverse lines remain straight and yet undergo a rotation. The plane cross section remain plane after bending. The bending moment causes the material within the bottom portion of the bar to stretch and the material within the top portion to compress. Consequently, between these two regions there must be a surface, called the neutral surface, in which longitudinal fibers of the material will not undergo any change in length.



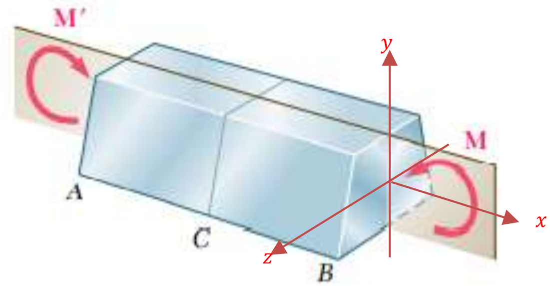
Before deformation



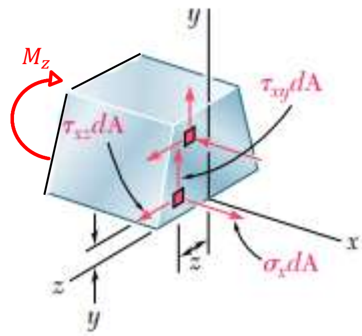
After deformation



Consider a prismatic beam with a cross section with vertical axis of symmetry subjected to pure bending moment in the plane of symmetry. Let us consider the internal loading by cutting the beam at C.



Consider a small elemental area dA on the cross section which is at a distance (y, z) from the y and z axes. The three stress components acting at that location can be considered as normal stress σ_{xx} and two shear stresses τ_{xy} and τ_{xz}.



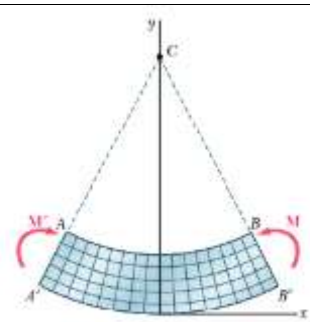
From the equilibrium consideration,

$\Sigma F_x = 0 \gg \int \sigma_{xx}\ dA = 0$	$\Sigma M_x = 0 \gg \int \tau_{xz}\ dA\ y - \int \tau_{xy}\ dA\ z = 0$
$\Sigma F_y = 0 \gg \int \tau_{xy}\ dA = 0$	$\Sigma M_y = 0 \gg \int \sigma_{xx}\ dA\ z = 0$
$\Sigma F_z = 0 \gg \int \tau_{xz}\ dA = 0$	$\Sigma M_z = 0 \gg -M_z - \int \sigma_{xx}\ dA\ y = 0$

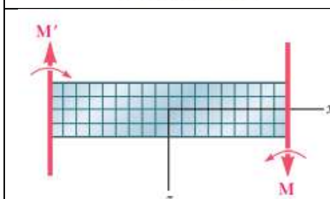
Let the member be divided in to a large number of small cubic elements whose faces are parallel to the co-ordinate planes. When the beam is subjected to pure bending (constant bending moment M), all the faces represented in the two projections are at $90°$ to each other. Hence, we can conclude that $\gamma_{xy} = \gamma_{xz} = 0$.

Therefore, $\tau_{xy} = \tau_{xz} = 0$.

The only non-zero stress component on the cross section is σ_{xx}. Hence, at any point of a slender member subjected to pure bending, we have a state of uni-axial stress state.

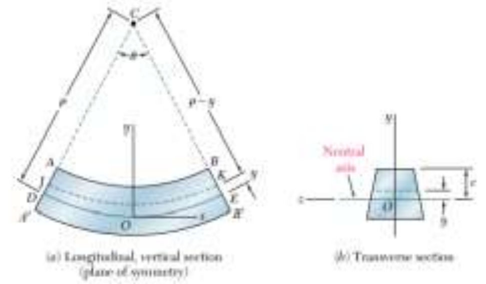


(a) Longitudinal, vertical section (plane of symmetry)



(b) Longitudinal, horizontal section

Referring to the figure, for a positive bending moment M (sagging moment), the line AB will have a decrease and $A'B'$ will have an increase in length. So, the stress σ_{xx} and strain ε_{xx} are negative (compressive) on the upper portion and positive (tensile) on the bottom part.



Therefore, there must be a surface parallel to the upper and lower surfaces, where σ_{xx} and ε_{xx} are zero. This surface is known as neutral surface and the line of intersection of the neutral surface with the cross section is known as neutral axis. Consider the origin of co-ordinates at the neutral surface.

Let ρ denotes the radius of arc DE, θ denotes the central angle corresponding to DE, and L denotes the undeformed length of DE.

$$L = \rho\theta$$

Consider the element JK situated at a distance y above the neutral surface. After bending, JK bends to an arc of length $L' = (\rho - y)\theta$.

The deformation of JK, $\delta = L' - L = (\rho - y)\theta - \rho\theta = -y\theta$.

The longitudinal strain on JK, $\varepsilon_{xx} = \frac{\delta}{L} = \frac{-y\theta}{\rho\theta} = \frac{-y}{\rho}$

The normal stress on the element JK, $\sigma_{xx} = E\varepsilon_{xx} = \frac{-E}{\rho} y$

$$\int \sigma_{xx}\, dA = 0 \ggg \int \frac{-E}{\rho} y\, dA = 0$$

$$\frac{-E}{\rho}\int y\, dA = 0 \ggg \int y\, dA = 0$$

$$A\bar{y} = 0 \ggg \bar{y} = 0$$

That is, the neutral axis passes through the centroid of the cross section

$$-M_z - \int \sigma_{xx}\, dA\, y = 0 \gg \int \sigma_{xx}\, y dA = -M_z$$

$$\int \frac{-E}{\rho} y\, y dA = -M_z \gg \frac{E}{\rho}\int y^2\, dA = M_z$$

$\int y^2\, dA = I_{zz}$, moment of inertia of the cross section of the beam w.r.t the z-axis (neutral axis).

$$\frac{EI_{zz}}{\rho} = M_z$$

$$\therefore \sigma_{xx} = \frac{-E}{\rho} y = -\frac{M_z}{I_{zz}} y$$

This is known as flexure formula.

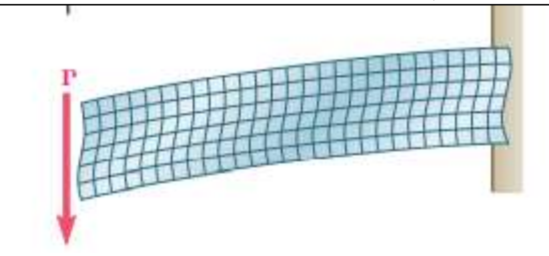
When $y = c$, (top surface) the stress is maximum.

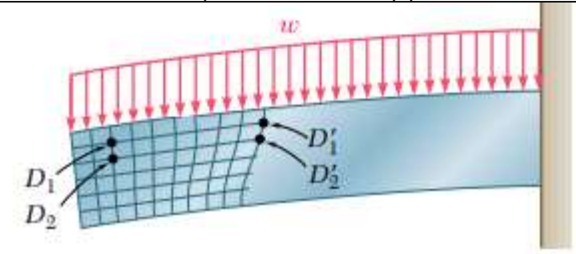
$\therefore \sigma_{xx}\,(\max) = -\frac{M_z}{\frac{I_{zz}}{c}} = -\frac{M_z}{S}$, S is known as the section modulus.

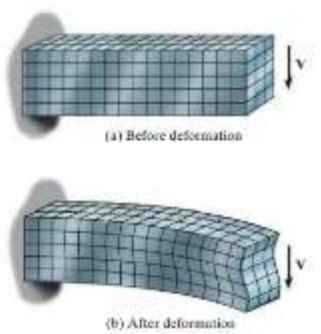
The bending stress is maximum at the top and bottom surfaces where y is maximum. One will be a compressive stress and the other will be a tensile stress depending upon the sign of the bending moment.

Even though, the results are derived for pure bending (constant bending moment and no shear force), it can be used for cases where shear forces are also present if the beam is slender (i.e., cross section is small compared to its length). Such beams are called Euler-Bernoulli beams (thin beams). Due to bending, cross sectional planes rotate and plane sections will remain plane after bending.

If cross section is large compared to the length, they are called Timoshenko beams (thick beams) and, in such beams, the effect of shear deformation cannot be neglected. Here, the cross-sectional planes will not remain plane after bending as shown below. In such cases, Timoshenko beam theory has to be applied.

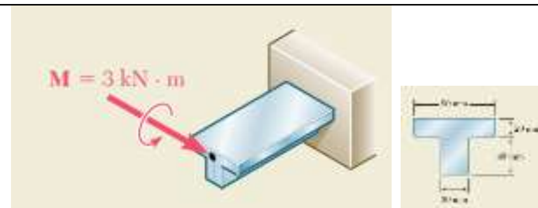


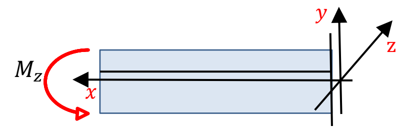




#Example 1

A cast iron machine part is acted upon by the $3\ kN.m$ couple as shown. Knowing that $E = 165\ GPa$ and neglecting the effect of fillets, determine (a) the maximum tensile and compressive streses in the casting, (b) the radius of curvature of the casting.

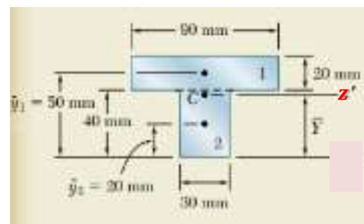




First, we have to locate the centroid of the cross section through which the neutral axis (z-axis) passes.

	A_i mm^2	y_i mm	A_iy_i
1.	$20\times 90 = 1800$	50	90000
2.	$40\times 30 = 1200$	20	24000
	$\Sigma A_i = 3000$		$\Sigma A_iy_i = 114000$

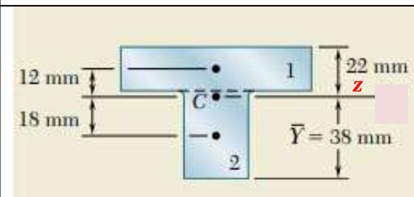
$$\bar{y} = \frac{\Sigma A_iy_i}{\Sigma A_i} = 38\ mm$$



Moment of inertia of the cross-section about the neutral axis (centroidal z-axis)

$$I_{zz} = \left\{\frac{90\times 20^3}{12} + 1800\times 12^2\right\} + \left\{\frac{30\times 40^3}{12} + 1200\times 18^2\right\}$$

$$I_{zz} = 868\times 10^3\ mm^4 = 868\times 10^{-9}\ m^4$$



The flexure formula is $\sigma_{xx} = -\frac{M_z}{I_{zz}}y$

Here, bending moment is negative as per the sign convention.

$$M_z = -3\ kN.m$$

The bending stress is maximum at the extreme surfaces $(y = +22\ mm\ and\ y = -38\ mm)$

$$\sigma_{xx_{y=+22mm}} = -\frac{(-3\times 10^3 N.m)}{868\times 10^{-9}\ m^4}(22\times 10^{-3}m) = +76\times 10^6\ Pa$$

$$\sigma_{xx_{y=-38mm}} = -\frac{(-3\times 10^3 N.m)}{868\times 10^{-9}\ m^4}(-38\times 10^{-3}m) = -131.6\times 10^6\ Pa$$

$$\sigma_{xx} = \frac{-E}{\rho}y = -\frac{M_z}{I_{zz}}y \ \ggg\ \frac{1}{\rho} = \frac{M_z}{EI_{zz}} = \frac{(-3\times 10^3 N.m)}{\left(165\times 10^9 \frac{N}{m^2}\right)(868\times 10^{-9}\ m^4)} = -20.95\times 10^{-3} m^{-1}$$

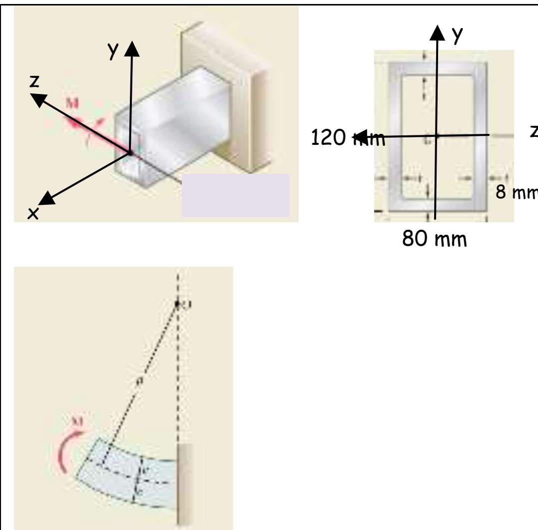
Negative sign shows that the curvature is convex. The radius of curvature,

$$\rho = 47.7\ m$$

#Example 2

The rectangular tube shown is extruded from an aluminium alloy for which $\sigma_Y = 150\ MPa, \sigma_U = 300\ MPa, E = 70\ GPa.$

Neglecting the effects of fillets, determine the maximum bending moment which can be applied considering a factor of safety of 3. Also determine the radius of curvature of the tube.



Since, the cross-section is symmetrical to both y and z axes, the centroid is at its centre. The neutral axis is the z-axis.

$$I_{zz} = \left\{\frac{80 \times 120^3}{12}\right\} - \left\{\frac{64 \times 104^3}{12}\right\} = 5521 \times 10^3\ mm^4 = 5521 \times 10^{-9}\ m^4$$

The allowable stress, $\sigma_{allow} = \frac{\sigma_U}{Factor\ of\ safety} = \frac{300\ MPa}{3} = 100\ MPa\ < \sigma_Y$

The allowable stress is less than the yield stress. Therefore, the tube remains within the elastic limit. Hence, the flexure formula is applicable.

The flexure formula is $\sigma_{xx} = -\frac{M_z}{I_{zz}} y$

Here, bending moment is positive as per the sign convention.

$$M_z = M$$

The bending stress is maximum at the extreme surfaces $(y = \pm 60\ mm)$

$$100 \times 10^6 \frac{N}{m^2} = -\frac{M}{5521 \times 10^{-9}\ m^4}(60 \times 10^{-3} m) \ggg M = 9201\ N.m$$

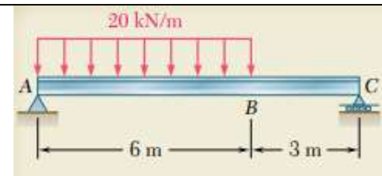
$$\sigma_{xx} = \frac{-E}{\rho} y = -\frac{M_z}{I_{zz}} y \ggg \frac{1}{\rho} = \frac{M_z}{EI_{zz}} = \frac{(9201 N.m)}{\left(70 \times 10^9 \frac{N}{m^2}\right)(5521 \times 10^{-9}\ m^4)} = +23.8 \times 10^{-3} m^{-1}$$

Negative sign shows that the curvature is concave. The radius of curvature,

$$\rho = 42\ m$$

#Example 3

The $W360 \times 79$ rolled-steel beam $(section\ modulus = 1270 \times 10^3 mm^3)$ AC is simply supported and carries the uniformly distributed load shown. Determine the location and magnitude of the maximum normal stress due to bending.

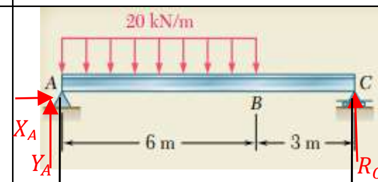


Considering the free-body diagram of the beam,

$$\Sigma F_x = 0 \ \ggg \ X_A = 0$$

$$\Sigma M_A = 0 \ggg \ R_C \times 9m - \ 20\frac{kN}{m} \times 6m \times 3m = 0 \gg \ R_C = 40\ kN$$

$$\Sigma F_y = 0 \ \ \ggg \ R_C + Y_A - 20\frac{kN}{m} \times 6m = 0 \ \ggg Y_A = 80\ kN$$



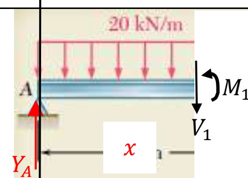
Section 1-1 $(0 \leq x \leq 6\ m)$

$$\Sigma F_y = 0 \ \ \ggg \ Y_A - V_1 - 20\frac{kN}{m} \times x = 0$$

$$V_1 = 80 - 20x\ kN \begin{Bmatrix} x = 0, \ V_1 = 80\ kN \\ x = 6m, V_1 = -40\ kN \end{Bmatrix}$$

$$\Sigma M_{1-1} = 0 \ \ \ggg \ -Y_A \times x + M_1 + 20\frac{kN}{m} \times \frac{x^2}{2} = 0$$

$$M_1 = 80x - 10x^2\ kNm \begin{Bmatrix} x = 0, & M_1 = 0 \\ x = 6m, & M_1 = 120\ kNm \end{Bmatrix}$$



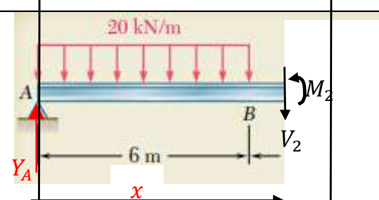
Section 2-2 $(6m \leq x \leq 9\ m)$

$$\Sigma F_y = 0 \ \ \ggg \ Y_A - V_2 - 20\frac{kN}{m} \times 6 = 0$$

$$V_2 = -40\ kN \begin{Bmatrix} x = 6m, \ V_2 = -40\ kN \\ x = 9m, V_2 = -40\ kN \end{Bmatrix}$$

$$\Sigma M_{2-2} = 0 \ \ \ggg \ -Y_A \times x + M_2 + 20\frac{kN}{m} \times 6 \times (x-3) = 0$$

$$M_2 = 80x - 120(x-3)\ kNm \begin{Bmatrix} x = 6m, & M_2 = 120\ kNm \\ x = 9m, & M_2 = 0 \end{Bmatrix}$$



The shear force becomes zero at D in section 1-1.

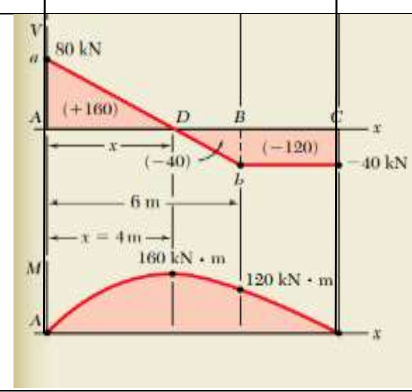
$$80 - 20x = 0 \ggg x = 4\ m$$

The maximum bending moment corresponding to point D

$$M_1 = 80x - 10x^2 = 80 \times 4 - 10 \times 4^2 = 160\ kN.m$$

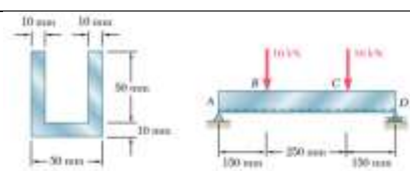
Neglecting the effect of shear deformation, the magnitude of maximum normal stress

$$\sigma_{xx} = \frac{M_{z\,max}}{S} = \frac{160 \times 10^3\ N.m}{1270 \times 10^{-6} m^3} = 126 \times 10^3\ Pa$$



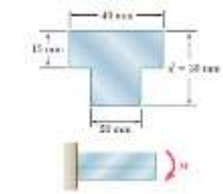
Exercise 1

Two vertical forces are applied to a beam of the cross section shown. Determine the maximum tensile and compressive stresses in portion BC of the beam



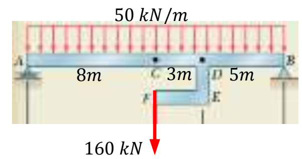
Exercise 2

The beam shown is made of nylon for which the allowable stress is 24 MPa in tension and 30 MPa in compression. Determine the largest moment M that can be applied to the beam



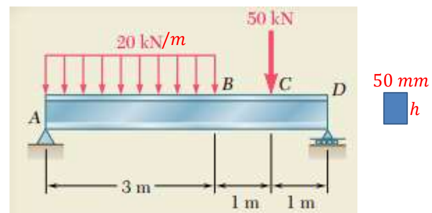
Exercise 3

A rigid steel bar DEF is welded to the straight steel beam AB of cross-section $50\ mm \times 50\ mm$. Draw the shear force and bending moment diagram of the beam AB for the load shown. Determine the maximum bending developed in the beam and its location.



Exercise 4

A 5m long, simply supported steel beam AD with rectangular cross-section is to carry loads as shown. The allowable normal stress for the grade of steel used is $160\ MPa$. If the width of the beam is $50\ mm$, what should be its depth h?



Exercise 5

Select a wide flanged beam for the above instead of rectangular section.

Shape	S, $mm^3 \times 10^3$
W410 × 38.8	629
W360 × 32.9	475
W310 × 38.7	547
W250 × 44.8	531
W200 × 46.1	451

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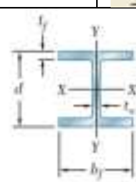
APPENDIX C Properties of Rolled-Steel Shapes
(SI Units)

W Shapes
(Wide-Flange Shapes)



Designation†	Area A, mm^2	Depth d, mm	Flange Width b_f, mm	Flange Thickness t_f, mm	Web Thickness t_w, mm	Axis X-X I_x $10^6\ mm^4$	Axis X-X S_x $10^3\ mm^3$	Axis X-X r_x mm	Axis Y-Y I_y $10^6\ mm^4$	Axis Y-Y S_y $10^3\ mm^3$	Axis Y-Y r_y mm
W920 × 449	57300	947	424	42.7	24.0	8780	18500	391	541	2560	97.0
201	25600	904	305	20.1	15.2	3250	7190	356	93.7	618	60.5
W840 × 299	38200	856	399	29.2	18.2	4830	11200	356	312	1560	90.4
176	22400	836	292	18.8	14.0	2460	5880	330	77.8	534	58.9
W760 × 257	32900	772	381	27.2	16.6	3430	8870	323	249	1310	86.9
147	18800	754	267	17.0	13.2	1660	4410	297	53.3	401	53.3
W690 × 217	27800	696	356	24.8	15.4	2360	6780	292	184	1040	81.3
125	16000	678	254	16.3	11.7	1190	3490	272	44.1	347	52.6
W610 × 155	19700	612	325	19.1	12.7	1290	4230	257	108	667	73.9
101	13000	602	228	14.9	10.5	762	2520	243	29.3	257	47.5
W530 × 150	19200	544	312	20.3	12.7	1010	3720	229	103	660	73.4
92	11800	533	209	15.6	10.2	554	2080	217	23.9	229	45.0
66	8390	526	165	11.4	8.89	351	1340	205	8.62	104	32.0
W460 × 158	20100	475	284	23.9	15.0	795	3340	199	91.6	646	67.6
113	14400	462	279	17.3	10.8	554	2390	196	63.3	452	66.3
74	9480	457	191	14.5	9.02	333	1460	187	16.7	175	41.9
52	6650	450	152	10.8	7.62	212	944	179	6.37	83.9	31.0
W410 × 114	14600	419	262	19.3	11.6	462	2200	178	57.4	441	62.7
85	10800	417	181	18.2	10.9	316	1510	171	17.9	198	40.6
60	7610	406	178	12.8	7.75	216	1060	168	12.0	135	39.9
46.1	5890	404	140	11.2	6.99	156	773	163	5.16	73.6	29.7
38.8	4950	399	140	8.76	6.35	125	629	159	3.99	57.2	28.4

References:

1. Mechanics of Materials, 2/e, J. M. Gere & S. P.Timoshenko, CBS Publishers.
2. Mechanics of Materials, 6/e, F.P. Beer & E. R. Johnston, McGraw Hill
3. Engineering Mechanics of Solids, 2/e, E. P. Popov, Pearson Education Asia.
4. Mechanics of Materials, 9/e, R. C. Hibbeler, Prentice Hall.